VIT-AP UNIVERSITY

CSE4006 – DEEP LEARNING

MECHANISTIC INTERPRETABILITY OF SAFETY NEURONS IN VISION-LANGUAGE MODELS

A structured literature review of probing, activation patching, and localized safety representations

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Abstract

Vision-language models (VLMs) can interpret both images and text, but they do not always make equally safe decisions across the two modalities. A harmful request may be refused when written plainly and handled differently when embedded in a picture. This review asks what happens inside the model when it makes these decisions. It compares research on safety neurons, internal circuits, probing, ablation, and activation-level interventions, rather than judging safety only from final answers.

Across the papers, there is no clear evidence for one universal “safety neuron” that controls every safe response. One study links text refusal to a relatively small group of neurons, while visual safety appears to involve a broader set of representational directions. Other work examines how safety-related circuits can be exploited in attacks and how internal interventions might reduce unsafe outputs. A useful distinction throughout this review is that finding information in an activation is not the same as proving that the information causes a model’s response.

The review uses a bounded, traceable identification record: 30 visible Google Scholar results captured across three queries and 16 retained research PDFs. Deduplicating their 46 appearances gives 38 distinct records. Eleven were excluded at title/abstract screening; eleven additional potentially relevant works were not present in the retained PDF corpus; and sixteen full texts were assessed. Eight VLM studies entered the core comparison. Because models and benchmarks differ, the results are synthesized qualitatively rather than pooled.

Keywords: vision-language models; mechanistic interpretability; safety neurons; activation patching; probing; activation steering; PRISMA 2020.

1. Introduction

Vision-language models combine an image encoder with language-processing components so that they can answer questions about pictures, read text in images, and respond to instructions that refer to visual details. These abilities make them useful, but they also introduce safety questions. For example, a model might refuse a harmful request when it is typed directly but respond differently when the request is embedded in an image. Recent work has started to examine whether safety behaviour follows the same internal path in both cases (Li et al., 2026; Fu et al., 2026).

A model’s final answer shows the outcome, but it does not explain how the model reached it. Mechanistic interpretability addresses this problem by examining internal activations and testing what happens when selected components are changed. The papers in this review use several approaches, including probing, neuron ablation, activation revision, and safety-subspace projection (Li et al., 2025; Li et al., 2026; Fu et al., 2026). These methods answer different questions, so their findings need to be interpreted according to the kind of evidence each method provides.

1.1 Problem statement

The problem addressed here is that the internal basis of safety in VLMs is still not fully understood. Safety behaviour might depend on a small number of neurons, a larger circuit, or features distributed across multiple directions. It may also differ depending on whether a request arrives as text or through an image. A related concern is how strong a causal claim can be made from each method. The reviewed studies use different models, harmful-content categories, benchmarks, and metrics, so a fair comparison must preserve those differences rather than treating the experiments as directly equivalent.

1.2 Scope and contribution

This review focuses on internal representations associated with safe and unsafe responses in vision-language models. It considers safety neurons, probing, activation patching, activation steering, and other causal interventions. Eight VLM studies from the supplied archive form the main comparison; two language-only papers are used only to explain related interpretability methods. The review does not claim to cover every publication in this growing area. Its contribution is a structured comparison of the available studies, with attention to the difference between identifying a representation and demonstrating that it has a causal role in model behaviour.

2. Objectives and Research Questions

The review was guided by four practical objectives:

- Identify the internal units used to analyse safety in VLMs, including neurons, layers, attention heads, circuits, and subspaces.
- Compare probing with interventions such as ablation, activation patching, steering, and causal mediation.
- Assess what the studies indicate about shared versus modality-specific safety mechanisms.
- Identify reproducibility, evaluation, and generalization gaps that limit deployment confidence.

RQ1. How are safety-related representations distributed across VLM components and input modalities?

RQ2. What can probing and activation-level interventions establish about the causal role of localized safety concepts?

RQ3. How do text and visual safety pathways overlap, and where do they appear to differ?

RQ4. Which evaluation and reporting gaps prevent reliable comparison and practical deployment?

3. Review Methodology

3.1 Reporting approach and information sources

Google Scholar was the documented discovery interface. Three query result pages were recorded on 18 October 2026, with their first ten visible entries retained for each query. On 28 October, I audited those 30 entries against the 16 PDFs previously collected. This is a bounded captured-results audit, not a claim to have exported or screened every result shown by Google Scholar.

Three Google Scholar query strings and the updated displayed interface counts supplied on 18 October 2026 are:

- S1: ("vision-language model" OR "multimodal large language model") AND (safety OR refusal OR jailbreak) AND (interpretability OR neuron OR circuit) — About 15,300 results displayed.
- S2: ("vision-language model" OR VLM OR MLLM) AND ("activation patching" OR probing OR ablation) AND (safety OR refusal) — About 12,700 results displayed.
- S3: ("safety neurons" OR "safety circuit" OR "safety subspace") AND ("vision-language model" OR multimodal) AND (causal OR intervention OR steering) — About 156 results displayed.
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The displayed totals for S1–S3 are approximate search-interface estimates; I did not use them as screening counts. Instead, the accompanying search evidence provides exactly 30 visible result appearances (10 per query). Two appearances represent repeated Scholar records, and six archive PDFs match search entries; these eight duplicate appearances were removed from the combined 46-item record.

That leaves 38 distinct records for a documented screening audit. Eleven were screened out from the available title/abstract evidence because they were surveys, attack-only evaluations or outside the narrow VLM safety-mechanism scope. Eleven more looked potentially relevant but had no PDF in the retained sixteen-paper archive, so I did not claim to have completed their full-text eligibility assessment. Sixteen available PDFs were evaluated in full; eight met the core inclusion rules.

3.2 Eligibility criteria

Criterion	Include in core synthesis	Exclude from core synthesis
Topic	Internal safety-related representations or interventions in VLMs/MLLMs	General safety benchmark or training work without internal analysis
Method	Empirical probing, ablation, patching, steering, causal analysis, or internal feature intervention	Output-only testing without internal representation analysis
Modality	Vision-language or multimodal language model	Language-only papers (contextual use only) or vision-only classifier studies
Evidence	Full-text PDF available and methods/outcomes described	Insufficient information or a primary task outside the review question
Date/language	English-language work dated 2020–2026	Outside date range or non-English full text

Some papers discuss VLM safety but do not examine the model's internal mechanisms. I kept the core review focused on studies that investigate safety-related representations or intervene on internal components. Two language-only studies were retained as background because they help explain the interpretability methods, but they are not treated as direct evidence about VLMs.

The eligibility rules covered English-language work published from 2020 to 2026, provided the full text was available and the methods and outcomes could be inspected. To enter the core synthesis, a paper had to study a vision-language or multimodal language model and examine an internal safety-related representation or intervene on a feature, neuron, layer, attention head, circuit, or subspace. Full-text conference papers, journal papers, and inspectable preprints could qualify. Language-only work was used for context only. Studies on general safety fine-tuning, safety-critical scene recognition, compositional failures, or vision-only classifiers were excluded when they did not address internal VLM safety mechanisms. No validated numerical risk-of-bias tool was used, and the studies were not pooled into a meta-analysis because their methods and outcomes differed.

3.3 Screening and data extraction

I reconciled the captured search-result titles and author clues against the retained PDFs, including alternate titles for the same work. For example, the "Precise Shield" listing and "Targeted Interpretable Safety Neuron Enhancement" PDF are two revisions of arXiv:2604.08881, not separate studies. The audit ledger records all 46 appearances and the identity of 38

unique records. Eleven title-level exclusions and eleven outside-archive candidates are listed individually. For the sixteen available papers, I read their titles, abstracts and relevant methods/results and documented each eligibility decision.

For each of the eight included studies I recorded the research objective, authors and year, method, models or benchmark, reported outcome, limitations and relevance to the research questions. This extraction is based on the full texts in the accompanying archive. Online records were also used to check selected bibliographic details. I did not infer missing numerical results or treat one paper's benchmark as directly interchangeable with another.

3.4 Quality and evidence appraisal

I appraised the papers using a small set of consistent questions: Is the model and intervention described clearly? Does the evaluation measure safety or refusal? Does the study check whether useful performance on benign tasks is preserved? Does it test new attacks or other models? And does it distinguish information that can be decoded from evidence that a component actually influences behaviour? This is a qualitative appraisal, not a validated risk-of-bias instrument, so I have not assigned numerical quality scores.

4. Results

4.1 Screening flow and corpus composition

Figure 1 shows what can actually be reconstructed from the two retained information sources. The log contains thirty visible Scholar entries; the supplied archive contains sixteen PDFs. There were eight duplicate appearances across these sources, giving thirty-eight distinct records. After eleven title/abstract exclusions, eleven potentially relevant results had no matching retained full text. Sixteen PDFs were assessed: eight were excluded from the core analysis (two retained only as language-model background and six outside the main scope) and eight were included.

Among the eight included VLM studies, two are dated 2025 and six 2026. This is a pattern in the retained eligible collection, not a measured publication trend for all research in the field. Several papers compare internal safety signals across modalities; others test probing, targeted ablation, activation revision, steering, or weight/subspace manipulation. Because task definitions and model families vary, method-level comparison is more reliable than ranking them by one reported accuracy or attack-success value.

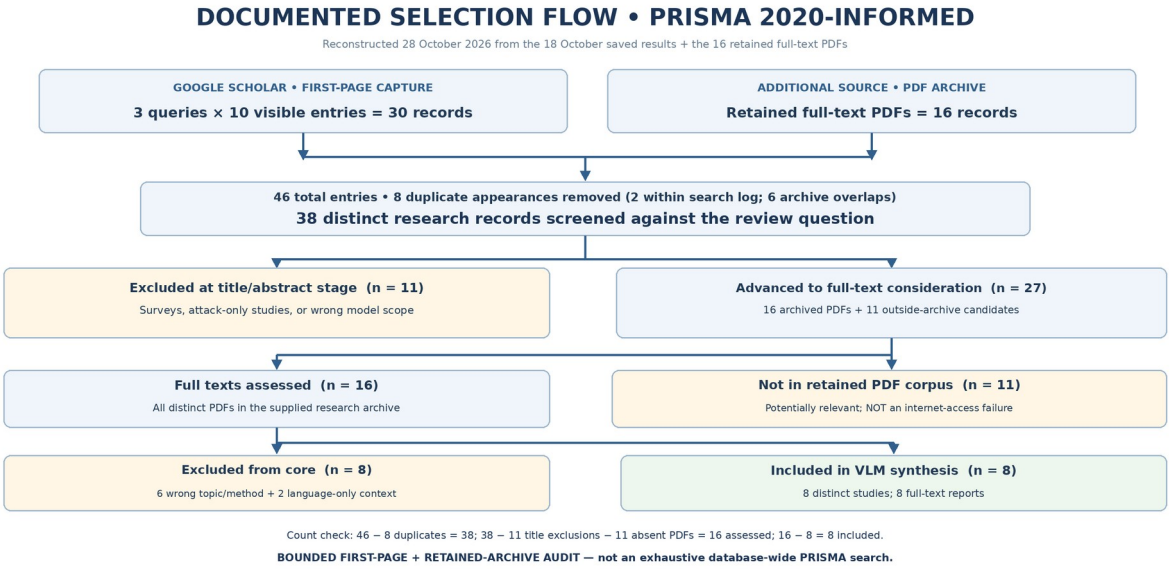


Figure 1. Actual countable records from the saved Google Scholar first-page capture and supplied-PDF archive (screening audit completed 28 October 2026). It is not an exhaustive database search.

The screening counts are linked to a supplementary 46-appearance identification ledger, a 38-record deduplicated decision list, and the existing file-by-file audit of sixteen PDFs. "Outside retained PDF corpus" does not mean the eleven articles were unavailable online; it means their full texts were not part of this review's archived evidence base.

Audited selection stage	Count	Interpretation
Records identified	46	Thirty saved Scholar entries and sixteen retained PDF entries
Duplicate appearances removed	8	Two within the Scholar listing; six overlaps with the PDF collection

Audited selection stage	Count	Interpretation
Unique records screened	38	All distinct titles screened using the captured bibliographic evidence
Screened out by title/abstract	11	Surveys, attack-only work or wrong model/mechanistic scope
Additional candidates without retained PDFs	11	Potentially relevant; full texts not part of the archived evidence set
Full-text PDFs assessed → core included	16 → 8	Eight exclusions: six outside core scope; two language-only context studies

4.2 Comparison of included studies

The table compares the eight core VLM studies using the information that could be supported by the available papers and verification records. The models and benchmarks differ, so a reported number in one paper is not automatically comparable with a number in another. Where an exact result was not re-extracted or a common metric was not available, I describe the reported finding rather than filling the gap with an estimate.

Study / year	Objective	Method / model	Dataset / metrics / reported finding	Limitation and research gap
Li et al. (2026), "Do VLMs Share Safety Neurons Across Modalities?"	Compare text and visual safety pathways across 10 VLMs.	Neuron-level localization, iterative ablation, modality-isolated ViSafe-Detect and ViSafe-Eval benchmarks.	Uses ViSafe-Detect and ViSafe-Eval across 10 VLMs. The paper reports about 88 text-safety neurons per model (<0.01% of neurons). Text refusal effects concentrate in about five GEVD directions, while visual effects grow through 50 directions under the tested intervention (Li et al., 2026).	The result is specific to tested models, prompts, and benchmarks. Visual safety is not shown to be governed by one universal neuron.
Xie et al. (2026), "Dissecting the Safety Circuit"	Study safety-related circuits and transferability of adversarial attacks.	Linear probing and Safety Circuit Intervention Attack (SCIA), which intervenes on safety and transferable circuits.	Evaluated on six open-source and three commercial VLMs using jailbreak settings including FigStep, MM-SafetyBench and Hades. The paper reports 59.7% average attack success rate (ASR) across evaluated open-source models. This is an attack-transfer result, not a defence success rate (Xie et al., 2026).	Attack-oriented evidence identifies a vulnerability, not a defence. It raises the need to test circuit overlap across architectures.
Fu et al. (2026), "Diagnosing and Repairing Unsafe Channels"	Identify unsafe internal channels and reduce unsafe behaviour while retaining multimodal capability.	Causal mediation and dual-modal safety-subspace projection (CARE).	Evaluates CARE on multiple safety benchmarks and reports improved safety robustness, transfer to unseen/adaptive attacks, and preservation of general multimodal capability. A directly comparable single score is not reproduced in this review because benchmark protocols differ.	Projection quality depends on how safety subspaces and benign/unsafe examples are constructed; matched utility metrics are needed.

Study / year	Objective	Method / model	Dataset / metrics / reported finding	Limitation and research gap
Gupta & Chakraborty (2026), "When Safety Overrides Vision"	Distinguish loss of visual grounding from refusal-oriented generation.	Hidden-state analysis and targeted activation-level intervention across VLMs and multimodal benchmarks.	Tests multiple VLM architectures and multimodal benchmarks. The paper reports that visual evidence remains influential during safety-induced abstention and that targeted activation intervention can restore grounded answering. The report does not quote a single pooled accuracy figure.	A restored answer is not automatically appropriate in every safety context. Harmless abstention and harmful compliance must both be measured.
Li et al. (2025), "Internal Activation Revision"	Reduce multimodal jailbreak vulnerability without updating model parameters.	Layer- and attention-head-level activation revision using revision vectors.	Reports average attack-success-rate reductions of 48.94% on SafeBench, 34.34% on Safe-Unsafe, 43.92% on Unsafe and 52.98% on MM-SafetyBench. These are the paper's reported benchmark-specific reductions; they should not be compared directly with another paper's ASR without matching protocols.	Performance depends on benchmark and revision-vector construction; safety gains must be evaluated alongside helpfulness and generalization.
Shi et al. (2026), "Targeted Interpretable Safety Neuron Enhancement…"	Address coupled multilingual and multimodal safety risks.	Neuron-level safety-neuron identification and targeted safety tuning.	Evaluates a neuron-targeted approach to coupled multilingual and multimodal safety. The paper reports improved safety while preserving general capabilities; specific benchmark-by-benchmark numbers are not reproduced here, so no numerical effect size is inferred.	Cross-lingual transfer and coverage of low-resource languages need careful independent evaluation.
Wang et al. (2025), "Steering Away from Harm"	Defend VLMs against image-based jailbreaks with inference-time intervention.	ASTRA: adaptive activation steering using transferable harmful-response directions.	Evaluates ASTRA against multimodal jailbreak attacks across multiple models and baselines. The paper reports resistance to several attack types with limited degradation on benign inputs. Exact comparable ASR figures are not reproduced in this table.	Steering directions can be prompt/model dependent; report should separate attack robustness from causal explanation of safety neurons.
D'Incà et al. (2026), "Safe Vision-Language Models via Unsafe Weights Manipulation"	Improve safety while preserving knowledge and safe-input behaviour.	SafeGround evaluation and activation-informed selection/negation of parameters associated with unsafe content.	Uses the SafeGround evaluation to examine safety alongside preservation of knowledge on safe inputs. The paper reports a favourable safety–knowledge-preservation trade-off against compared methods; this is adjacent to, but not the same as, a neuron-level causal test.	Weight manipulation is adjacent to, but not identical with, neuron-level causal explanation; evaluation on benign inputs is essential.

4.3 Contextual papers not counted in the core synthesis

Chen et al. (2025) examine safety neurons in language-only large language models using activation contrasting and dynamic activation patching. Their paper reports that roughly 5% of neurons could be identified as safety neurons and that patching restored more than 90% of safety performance on the red-teaming benchmarks they evaluated (Chen et al., 2025).

These results belong to the models and tests in that paper; they do not show that VLMs use the same proportion of neurons or the same mechanism. Safe-SAIL examines sparse-autoencoder features for fine-grained safety analysis in language models (Weng et al., 2026). It is useful methodological context, but it is not direct evidence about VLMs.

5. Cross-study Synthesis and Discussion

5.1 Localization is not the same as a universal safety switch

A clear difference in the reviewed evidence is the contrast between text and visual safety. Li et al. report that, in the models they tested, text refusal could be linked to a relatively small set of neurons, whereas visual safety was spread across more representational directions (Li et al., 2026). The study compares ten VLMs and separates text and visual evaluations, but the result still depends on the tested architectures, prompts, and definition of refusal. It would be premature to conclude that one fixed group of neurons controls safety in every model and situation.

For this reason, the phrase “safety neuron” should be used with care. What looks localized can depend on the task, the layer being examined, and the number of components changed during an intervention. A probe may show that safety-related information is present in an activation without proving that the model uses it to decide whether to refuse. Conversely, changing a component may affect the answer without that component representing safety alone. Strong claims should state which model, modality, behaviour, and causal test they refer to.

5.2 Probing, patching, ablation, and causal intervention

Probing asks whether information about safety can be recovered from a model’s internal activations. It is useful for locating candidate features, but a successful probe does not by itself show that the model relies on those features. Ablation and activation patching go further by changing internal components and observing whether the output changes. Even these interventions need controls: other pathways may carry similar information, the model may compensate for the change, or the intervention may damage general capability. The reviewed papers illustrate different levels of intervention, from neuron ablation to layer/head activation revision and dual-modal safety-subspace projection (Li et al., 2025; Li et al., 2026; Fu et al., 2026).

The study by Xie et al. also shows why it is important to distinguish understanding a circuit from using it as a defence. Their Safety Circuit Intervention Attack (SCIA) uses probe-identified circuits to help adversarial images transfer to black-box VLMs. The paper reports a 59.7% average attack success rate across the evaluated open-source models (Xie et al., 2026). This is an attack result, not a measure of defence effectiveness. In contrast, CARE, Internal Activation Revision, ASTRA, and unsafe-weights manipulation study interventions intended to improve safety. A more direct comparison would test these methods against the same attacks, on held-out models, while also measuring performance on benign inputs.

5.3 Safety must be evaluated with usefulness and grounding

A model that refuses every difficult image question might appear safe on a narrow harmful-compliance metric, but it would also be less useful. Gupta and Chakraborty report that visual information can remain influential even when safety alignment leads the model to abstain, and they examine targeted activation interventions (Gupta & Chakraborty, 2026). D’Incà et al. also consider whether safe behaviour can be improved while useful knowledge is retained (D’Incà et al., 2026). A balanced evaluation should measure harmful compliance and appropriate refusal separately, alongside over-refusal, visual grounding, ordinary task performance, and robustness to new attacks.

5.4 What the current evidence can and cannot establish

Taken together, the papers suggest that internal activations are relevant to multimodal safety and that interventions on those activations can change model behaviour (Li et al., 2025; Fu et al., 2026; Wang et al., 2025). However, the core VLM studies provide more direct evidence about neuron ablation, probing, steering, causal mediation, and safety-subspace projection than about conventional activation patching. The activation-patching paper in the supplied archive studies a vision-only classifier, so it helps explain the method but is not direct evidence about VLM safety. In this review, activation patching is therefore discussed as methodological background rather than treated as a major source of VLM-specific findings. The available studies do not establish a single safety mechanism that works across all architectures and tasks.

6. Research Gaps and Future Directions

6.1 Matched cross-modal causal experiments

One useful next step is to build matched examples in which the same harmful request is presented as plain text, as text embedded in an image, and as a visual scene. Researchers could then apply the same probing, patching, and ablation procedures to each version. This would make it easier to test whether the same internal units are involved across modalities.

Benign image questions and ambiguous visual examples should be included as controls so that the intervention is not rewarded simply for making the model refuse more often.

6.2 Standardized reporting of interventions

Reproducibility would improve if papers consistently reported the model checkpoint, the exact layers or components changed, intervention strength, how activation vectors were constructed, calibration examples, prompt templates, decoding settings, and computational cost. Studies that identify individual neurons should also state the selection threshold, check whether the model repairs or routes around the intervention, report how many units were tested, and show whether the result holds across random seeds. Public code and shared extraction tables would make it easier to reproduce and compare results.

6.3 Safety–utility evaluation with independent judges

A shared evaluation suite should measure harmful compliance and over-refusal separately, then report visual question answering, grounding, helpfulness, and answer correctness as well. Automatic safety scores are useful for scale, but independent or blinded human review could identify cases where refusal is appropriate for one prompt and excessive for another. This matters because an intervention can reduce unsafe answers while also blocking harmless questions.

6.4 Generalization across models, attacks, and languages

Future interventions should be tested on model families that were not used during development, as well as on unfamiliar attack styles, multilingual prompts, and low-resource languages. A feature direction found in one model should not be assumed to work in another without testing. The transfer experiments reported for SCIA and the multilingual goals of targeted safety-neuron enhancement both make generalization an important part of the problem.

6.5 Mechanism-level robustness and independent replication

A safety intervention should be tested against adaptive attackers who know how the intervention works. Researchers should check whether it can be bypassed, whether it suppresses harmless visual concepts, and whether it introduces other failures. Independent replication using open checkpoints and public evaluation scripts would help show whether a result holds beyond the benchmark on which the method was first tested.

7. Threats to Validity and Limitations

The major limitation is selection coverage. Although the revised audit gives real counts for the 30 captured Scholar entries and 16 saved PDFs, each query contributed only its first ten visible results. We did not preserve complete database exports or screen all 15,300/12,700/156 approximate search hits. Eleven potentially relevant works found in the captured results also lacked copies in the retained PDF corpus; some have public full texts, so the review could miss important studies. The flow is reproducible for this bounded set, not a comprehensive PRISMA search of the field.

The eight-study synthesis consequently remains narrow and may suffer from availability and selection bias. Differences in model families, visual encoders, prompt design, harm categories, attack strength and score definitions also complicate comparison. The study-quality assessment is qualitative rather than an independent risk-of-bias tool, and the review does not attempt meta-analysis.

Mechanistic claims additionally depend on the task and intervention. Showing that a probe predicts a refusal does not establish causality, while an intervention that changes outputs on one dataset does not establish a universal safety switch. The screening ledger, PDFs and eligibility reasons make the supported claims inspectable, but independent replication and a full multi-database retrieval would be needed for a definitive systematic review.

8. Conclusion

This review examined how mechanistic interpretability is being used to study safety-related representations in vision-language models. The eight core studies examine neurons, circuits, layers, attention heads, and feature subspaces, and several intervene on these components to investigate or reduce unsafe behaviour (Li et al., 2025; Xie et al., 2026; Fu et al., 2026). The evidence supports internal activation analysis as a useful research approach, but it does not show that one small, universal set of safety neurons controls all multimodal safety behaviour.

The next step should not be limited to finding more candidate neurons. Researchers need to compare equivalent requests presented through text and images, intervene on the internal features they identify, and check whether the model can still answer harmless visual questions correctly. That would help distinguish a feature that merely correlates with refusal from one that has a causal and useful role in safety.

The main practical point is that a good safety intervention should reduce harmful compliance without making the model unnecessarily refuse harmless visual questions. Matched text-and-image prompts, controlled interventions, and tests on models that were not used during development would make claims about safety-related neurons more convincing.

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Appendix A. Search strings and screening record

Identification sources were the preserved first-page Google Scholar results (three queries, ten entries each) and the sixteen-PDF archive assembled during the review. The information-source table below distinguishes those auditable records from secondary publisher or arXiv pages consulted to verify selected references. No other database is claimed to have been searched systematically.

Source	Date recorded	Purpose in this report	Count status
Google Scholar	18 Oct 2026 capture; 28 Oct 2026 audit	S1–S3 first ten visible Google Scholar results per query; deduplicated in screening ledger.	30 captured entries; 2 duplicate search appearances; 28 unique Scholar records
Publisher / proceedings pages	17–28 Oct 2026	Used only for selected bibliographic checks and paper-version identity matching.	No systematic independent publisher-database query performed
arXiv	17 Oct 2026	Targeted checks of preprint metadata and paper text where relevant.	No systematic query log or result count; not counted as a database search.
Other bibliographic databases	Not documented	No systematic search record retained for IEEE Xplore, Scopus, Web of Science, PubMed, ACM Digital Library, SpringerLink, or ScienceDirect.	No search or screening counts claimed.

The original queries and result-page excerpts are dated 18 October 2026. Their record-level deduplication and screening audit was completed on 28 October 2026; the results and reasons appear in the supplementary CSV files. The three Scholar query-level approximate result totals are not record counts; only the 30 individual displayed entries were used.

Search ID	Exact query	Date and filters	Recorded interface count
S1	("vision-language model" OR "multimodal large language model") AND (safety OR refusal OR jailbreak) AND (interpretability OR neuron OR circuit)	18 Oct 2026; Google Scholar; Articles; relevance sort; date control displayed “Any time” (as reported in supplied capture)	~15,300 (approximate interface estimate; not screened records)
S2	("vision-language model" OR VLM OR MLLM) AND ("activation patching" OR probing OR ablation) AND (safety OR refusal)	18 Oct 2026; Google Scholar; Articles; relevance sort; date control displayed “Any time” (as reported in supplied capture)	~12,700 (approximate interface estimate; not screened records)

Search ID	Exact query	Date and filters	Recorded interface count
S3	("safety neurons" OR "safety circuit" OR "safety subspace") AND ("vision-language model" OR multimodal) AND (causal OR intervention OR steering)	18 Oct 2026; Google Scholar; Articles; relevance sort; date control displayed “Any time” (as reported in supplied capture)	~156 (approximate interface estimate; not screened records)

Selection arithmetic: 30 captured Scholar entries plus 16 retained PDFs = 46 record appearances; eight duplicate appearances (two within the Scholar capture and six overlapping the PDF archive) leave 38 unique research records. Of those, eleven were excluded by title/abstract and eleven remained potentially relevant without a corresponding retained PDF. The sixteen actual PDF full texts were assessed, with eight core inclusions, two language-only context sources, and six excluded for topic or method.

Eligibility: English-language research published during 2020–2026; inspectable full-text empirical studies of internal safety representations or interventions in VLMs/MLLMs for core inclusion. The search-result screening used title and available bibliographic clues; a candidate was not declared ineligible simply because no PDF had been retained. The eleven such cases are explicitly labelled as unresolved full-text appraisal, which limits completeness.

No.	Paper title	Author/year as available	Decision and reason
1	Do VLMs Share Safety Neurons Across Modalities?	Li et al. (2026)	Include — core VLM safety-neuron analysis
2	Dissecting the Safety Circuit	Xie et al. (2026)	Include — probing and safety-circuit intervention
3	Diagnosing and Repairing Unsafe Channels	Fu et al. (2026)	Include — causal analysis and dual-modal subspaces
4	When Safety Overrides Vision	Gupta & Chakraborty (2026)	Include — refusal dynamics and activation intervention
5	Internal Activation Revision	Li et al. (2025)	Include — VLM activation revision for safety
6	Targeted Interpretable Safety Neuron Enhancement for Multilingual VLLMs	Shi et al. (2026)	Include — neuron-level multilingual/multimodal safety tuning
7	Steering Away from Harm	Wang et al. (2025)	Include — adaptive activation steering defence
8	Safe Vision-Language Models via Unsafe Weights Manipulation	D’Incà et al. (2026)	Include — activation-informed intervention and safety–utility evaluation
9	Towards Understanding Safety Alignment	Chen et al. (2025)	Context only — language-only LLM study; excluded from core count
10	Safe-SAIL	Weng et al. (2026)	Context only — language-only SAE feature analysis
11	Do VLMs Have Bad Eyes?	Vaithinathan Aravindan et al.	Exclude — mechanistic interpretability, but compositionality rather than safety
12	Do We Really Need Curated Malicious Data for Safety Alignment?	Wang et al.	Exclude — multimodal safety fine-tuning/data study without internal interpretability focus
13	Hyperbolic Safety-Aware Vision-Language Models	Poppi et al.	Exclude — safety-aware representation/retrieval, not the defined internal safety mechanism focus
14	ScVLM: Enhancing VLM for Safety-Critical Event Understanding	Shi et al.	Exclude — safety-critical scene understanding, not model safety alignment
15	TAB: Transformer Attention Bottlenecks	Rahmanzadehgervi et al.	Exclude — visual debugging/intervention, not safety-neuron analysis
16	What Can Activation Patching Tell Us About Adversarial Examples?	Rathi & Yu	Exclude — vision-only classifier rather than VLM safety

Flow interpretation: this is a bounded identification-to-inclusion audit of thirty individually captured search results and sixteen retained PDF reports. It reports actual recorded duplicate and screening counts without pretending that all Google Scholar result pages were downloaded. The eleven otherwise potentially eligible records outside the retained PDF corpus are a missing-evidence limitation, not evidence that full texts could not be found online.

Appendix B. Interpretation notes

The audit includes all 46 record appearances, links them to 38 deduplicated identities and records reasons for each candidate disposition. This is a newly reconstructed auditable search slice, not a retroactive claim that the full Google Scholar results were screened during the original 18 October search. Findings and exact metrics are attributed only to the eight core full texts or verified proceedings records.

The included research-paper ZIP contains the 16 documents listed in Appendix A. Some files were compressed for the course upload size limit; text and page order were preserved, while image quality may be reduced in the compressed copies. The original uncompressed archive should be retained separately if available.

Review status: a PRISMA 2020-guided structured review with a transparent, limited captured-results selection protocol. Its record-level flow is valid for the saved first pages and supplied archive, but a fully exhaustive systematic review would require complete searches, exports and full-text checks for the eleven unresolved candidates. No registered protocol or meta-analysis was used.

Appendix C. Qualitative study-appraisal matrix

The matrix records whether each included paper addresses key appraisal questions. “Reported” means the paper describes the element in its method or publisher summary; it does not imply independent replication. No numeric quality score is assigned because the review did not apply a validated risk-of-bias instrument.

Study	Internal unit / method stated	Intervention evidence	Safety–utility check	Generalization evidence	Appraisal caution
Li et al. (2026)	Neuron-level analysis and iterative ablation	Yes: targeted ablation	Capability evaluated; detailed figures not extracted here	Multiple VLMs and modality-isolated benchmarks	Intervention results depend on the tested models and safety definitions
Xie et al. (2026)	Probe-identified safety and transferable circuits	Yes: circuit intervention for attack	Attack performance; benign utility is not the primary aim	Transfer to unseen black-box VLMs reported	Attack-focused; does not establish a defence mechanism
Fu et al. (2026)	Causal mediation and dual-modal subspaces	Yes: activation projection	General multimodal capability reported	Unseen/adaptive attack transfer reported	Projection construction and matched metric reporting matter
Gupta & Chakraborty (2026)	Hidden-state decoding dynamics	Yes: refusal-related intervention	Grounded answering considered	Multiple architectures and benchmarks reported	Must distinguish appropriate refusal from over-refusal
Li et al. (2025), IAR	Layer/head-level revision vectors	Yes: activation revision	Helpfulness described as minimally affected	Multiple benchmarks reported	Effect size depends on benchmark and vector construction
Shi et al. (2026)	Neuron-level multilingual/multimodal safety tuning	Yes: targeted tuning	Needs joint safety and language capability evaluation	Cross-language and cross-modal scope is explicit	Low-resource-language coverage needs independent testing
Wang et al. (2025), ASTRA	Transferable harmful-response steering vectors	Yes: adaptive activation steering	Small benign-input performance drop reported	Unseen attack types reported	Adaptive attackers and model transfer remain important tests
D’Incà et al. (2026), UWM	Activation-informed parameter selection	Yes: parameter manipulation	Safe-input knowledge preservation is central	Multiple safety comparisons reported	Not identical to neuron-level causal localization

Three patterns emerge from this appraisal. First, the included studies generally go beyond output-only safety testing: they inspect internal states and several perform an intervention. However, the causal strength of these interventions varies. Probing identifies decodable information, while ablation, patching, steering, and projection test behaviour under a manipulation; each still requires controls to rule out collateral effects and redundant pathways.

Second, safety–utility evaluation is uneven. Some papers explicitly report benign-input performance, helpfulness, or preservation of general multimodal capability, whereas attack-focused circuit analysis is primarily designed to understand or improve adversarial transfer. This difference matters when interpreting the results: a method that increases refusal or attack resistance is not automatically better if it also blocks harmless image questions.

Third, generalization is a recurring open issue. Several studies report evaluation across multiple models, benchmarks, or unseen attacks, but the scope of those tests differs. A stronger future appraisal would require common held-out models, standardized attack suites, blinded human review, and a shared set of benign grounding tasks. These criteria would make it easier to distinguish robust safety mechanisms from interventions tuned to a particular model or benchmark.

Appendix D. Review development record (12–18 October 2026)

Development audit: topic and preliminary reading were undertaken during 12–18 October 2026. Three Scholar result pages supplied on 18 October preserved ten entries each. On 28 October, the visible records were consolidated with the sixteen retained PDFs, duplicate/version identities were reconciled and a new 46-appearance screening ledger was created. These stages must not be confused with a previously completed database-wide screen.

- **12 October —** Defined the topic around mechanistic interpretability of safety-related representations in vision-language models and began framing the problem.
- **13 October —** Refined the scope, review objectives and research questions; distinguished internal safety analysis from general VLM safety evaluation.
- **14 October —** Organized the candidate-paper collection and began reading abstracts and relevant method/result sections against the inclusion criteria.
- **15 October —** Compared the methods used across the papers, including probing, neuron ablation, activation patching, steering and safety-subspace interventions.
- **16 October —** Drafted the cross-study synthesis, limitations and future research directions; checked which numerical claims could be supported by the available papers.
- 28 October — Reconciled 30 captured Google Scholar entries with the sixteen retained full-text PDFs, verified the HiddenDetect publication/preprint match and the “Precise Shield”/“Targeted Interpretable” version match, recorded eight duplicate appearances and screened 38 unique titles against the defined scope.

Counts note: the Google Scholar interface estimates are not used in the PRISMA figure. The auditable unique-record ledger contains 38 items. Eleven screened-out titles and eleven potentially relevant papers without archived full texts are documented as separate cases; no online unavailability is asserted.

Provenance: the 18 October Search Evidence Log preserves the three queries and thirty pasted result entries. The 28 October Identification and Screening Ledger gives each original result appearance a canonical study ID, an outcome and an explanation. The original sixteen-file Archive Screening Record documents the subsequent full-text decisions.

Appendix E. Mandatory study-comparison extraction matrix

This matrix separates the required extraction fields. Dataset and metric descriptions are reported only to the level supported by the current review record; where an exact comparable numerical result was not re-extracted, this is stated rather than inferred. Metrics from different benchmarks are not directly comparable.

No.	Author / year	Method / model	Dataset / benchmark	Metrics / key finding	Limitation / research gap
1	Li et al. (2026)	Neuron localization; iterative ablation	ViSafe-Detect and ViSafe-Eval; 10 VLMs	About 88 text-safety neurons per model (<0.01%); text effects concentrated in ~5 GEVD directions, visual effects spread across up to 50 tested directions.	Model, prompt and safety definitions may limit generalisation; no universal safety neuron established.
2	Xie et al. (2026)	Linear probing; SCIA circuit intervention	FigStep, MM-SafetyBench, Hades; six open-source and three commercial VLMs evaluated	59.7% average attack success rate across evaluated open-source models; attack result, not defence success.	Attack-oriented; needs matched benign-utility and cross-architecture evaluation.
3	Fu et al. (2026)	Causal mediation; dual-modal safety-subspace projection (CARE)	Multiple safety benchmarks	Reports improved safety robustness and transfer to unseen/adaptive attacks while preserving general multimodal capability; no pooled score reproduced.	Subspace construction and matched utility metrics require further scrutiny.
4	Gupta & Chakraborty (2026)	Hidden-state analysis; targeted activation intervention	Multiple VLM architectures and multimodal benchmarks	Reports visual evidence remains influential during safety-induced abstention; intervention can restore grounded answering.	Must distinguish appropriate refusal from harmful compliance and over-refusal.
5	Li et al. (2025)	Layer/head activation revision vectors	SafeBench, Safe-Unsafe, Unsafe, MM-SafetyBench	Reported ASR reductions: 48.94%, 34.34%, 43.92%, and 52.98%, respectively.	Benchmark-specific results; utility and generalisation need matched protocols.
6	Shi et al. (2026)	Neuron-level identification and targeted safety tuning	Multilingual/multimodal VLM settings	Reports improved safety while preserving general capabilities; no common effect size inferred.	Low-resource-language coverage and independent cross-lingual validation remain needed.
7	Wang et al. (2025)	ASTRA adaptive activation steering	Multimodal jailbreak attacks across models and baselines	Reports resistance across several attack types with limited benign-input degradation; exact comparable ASR figures not reproduced.	Steering may be prompt/model dependent; test adaptive attackers and separate defence from mechanism explanation.
8	D'Incà et al. (2026)	Activation-informed selection/negation of unsafe parameters (UWM)	SafeGround and comparisons with other methods	Reports a favourable safety–knowledge-preservation trade-off; no pooled figure inferred.	Adjacent to, but not identical with, neuron-level causal localization.

Interpretation note: the table standardises the assignment's requested fields but does not imply that studies used the same benchmark, metric definition, model checkpoint, or evaluation protocol. Numerical values must be read in their original paper-specific context.

Appendix F. Supplied-PDF and reference verification audit

The 16 PDFs in the supplied research archive were inventoried by filename and checked against the paper title/author block visible in the PDF. The decisions below follow the report's eligibility criteria. "Context only" means the work is not counted as direct VLM evidence. This audit verifies the supplied collection; it does not substitute for a search-result export or prove that the collection captures all relevant literature.

Title verified from supplied PDF	Author/year used in report	Audit decision / note
Diagnosing and Repairing Unsafe Channels in Vision-Language Models via Causal Discovery and Dual-Modal Safety Subspace Projection	Fu et al. (2026)	Core inclusion
Dissecting the Safety Circuit: Neuronal Intervention for Transferable Adversarial Attacks on VLMs	Xie et al. (2026)	Core inclusion
Do VLMs Share Safety Neurons Across Modalities?	Li et al. (2026)	Core inclusion
When Safety Overrides Vision: Exploring Dynamics between Vision Influence and Safety Alignment in Vision-Language Models	Gupta & Chakraborty (2026)	Core inclusion
Internal Activation Revision: Safeguarding Vision Language Models Without Parameter Update	Li et al. (2025)	Core inclusion; seven-author list checked against AAAI proceedings metadata
Targeted Interpretable Safety Neuron Enhancement for Multilingual Vision-Language Large Models	Shi et al. (2026)	Core inclusion
Steering Away from Harm: An Adaptive Approach to Defending Vision Language Model Against Jailbreaks	Wang et al. (2025)	Core inclusion
Safe Vision-Language Models via Unsafe Weights Manipulation	D'Incà et al. (2026)	Core inclusion
Towards Understanding Safety Alignment: A Mechanistic Perspective from Safety Neurons	Chen et al. (2025)	Context only; language-only
Safe-SAIL: Towards a Fine-grained Safety Landscape of Large Language Models via Sparse Autoencoder Interpretation Framework	Weng et al. (2026)	Context only; language-only
Do VLMs Have Bad Eyes? Diagnosing Compositional Failures via Mechanistic Interpretability	Vaithinathan Aravindan et al. (2025)	Excluded from core; compositionality rather than safety alignment
Do We Really Need Curated Malicious Data for Safety Alignment in Multi-modal Large Language Models?	Wang et al. (2025)	Excluded from core; safety-data/training focus without the required internal-mechanism analysis
Hyperbolic Safety-Aware Vision-Language Models	Poppi et al. (2025)	Excluded from core; safety-aware representation/retrieval rather than defined internal safety mechanism
ScVLM: Enhancing Vision-Language Model for Safety-Critical Event Understanding	Shi et al. (2025)	Excluded from core; safety-critical scene understanding rather than model safety alignment
TAB: Transformer Attention Bottlenecks Enable User Intervention and Debugging in Vision-Language Models	Rahmanzadehgervi et al. (2025)	Excluded from core; visual debugging/intervention not specifically safety-neuron analysis
What Can Activation Patching Tell Us About Adversarial Examples?	Rathi & Yu (2024)	Excluded from core; vision-only classifier, not VLM safety

Bibliographic verification note. The authoritative AAAI proceedings record lists seven authors for "Internal Activation Revision": Qing Li, Jiahui Geng, Derui Zhu, Zongxiong Chen, Kun Song, Lei Ma, and Fakhri Karray. The reference entry above now includes Derui Zhu. The author list should be checked against the proceedings record rather than inferred from an abbreviated author block.

Supplementary audit files: (1) Google Scholar Search Evidence Log (30 captured appearances); (2) PRISMA Identification and Screening Ledger (46 appearances); (3) PRISMA Unique Record Decisions (38 studies); (4) Archive Screening Record (16 PDF decisions); and (5) standalone PRISMA count diagram. The search remains bounded to captured results rather than all Google Scholar hits.